

Data Encryption Standard (DES)

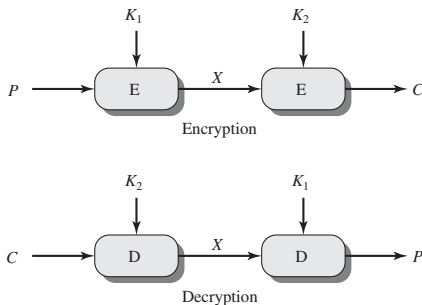
- The most widely used encryption is based on the Data Encryption Standard (DES) adopted in 1977 by the National Institute of Standards and Technology (NIST), USA.
- For DES, data are encrypted in 64-bit blocks using a 56-bit key.
- The encryption algorithm transforms 64-bit input in a series of steps into a 64-bit output.
- The same steps, with the same key, are used to reverse the encryption (decryption).
- Mathematically, $DES : \{0, 1\}^{64} \times \{0, 1\}^{56} \longrightarrow \{0, 1\}^{64}$ such that the ciphertext be $C = DES_K(P)$, where $K \in \{0, 1\}^{56}$ is the 56-bit key, $P \in \{0, 1\}^{64}$ is the plaintext message (block) and $C \in \{0, 1\}^{64}$ is the ciphertext block.

Data Encryption Standard (DES)

- DES finally and definitely proved insecure in July 1998, when the Electronics Frontier Foundation (EFF) announced that it had broken a DES encryption using a special-purpose “DES cracker” machine that was built for less than 250,000 USD.
- The attack took less than three days.

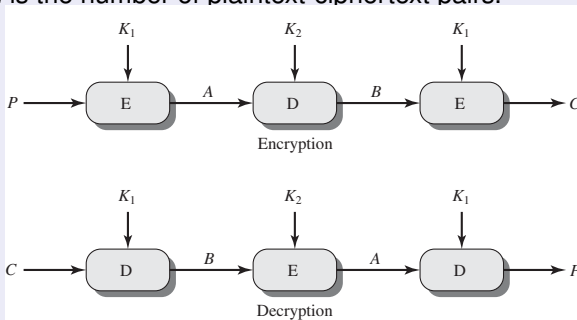
Double DES (2DES)

- It uses two 56-bit keys K_1 and K_2 , and 64-bit plaintext block.
- It produces 64-bit ciphertext block.
- Known-plaintext attack (meet-in-the-middle attack) is possible against 2DES to derive two keys K_1 and K_2 , which has a key size of 112 bits and with an effort on the order of 2^{56} .



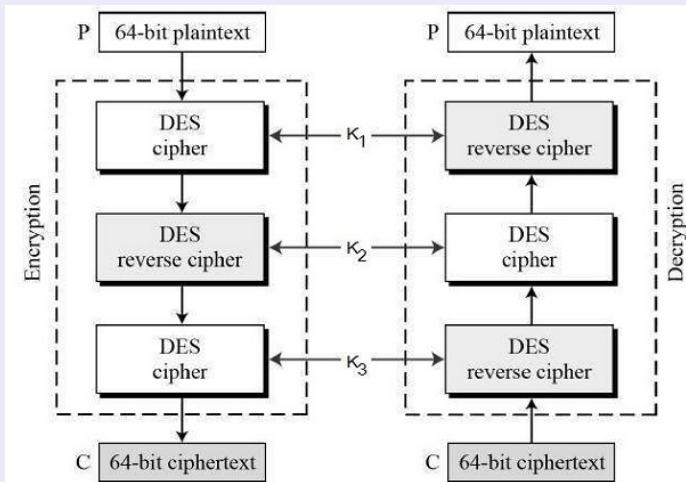
Triple DES with Two Keys (3DES with Two Keys)

- It uses two 56-bit keys K_1 and K_2 , and 64-bit plaintext block.
- It produces 64-bit ciphertext block.
- It is also vulnerable to known-plaintext attack (meet-in-the-middle attack) to derive two keys K_1 and K_2 .
- The expected running time of this attack is on the order of $2^{120 - \log_2 n}$, where n is the number of plaintext-ciphertext pairs.





Triple DES with Three Keys (3DES with Three Keys)



K_1 , K_2 and K_3 : three 56-bit keys

Online Demo on DES Encryption and Decryption

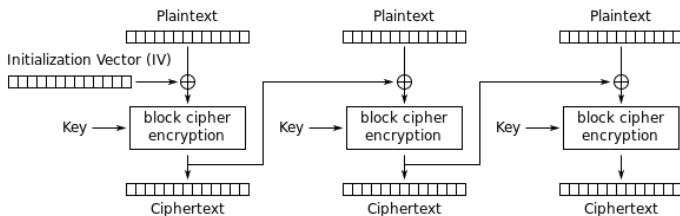
- Generating parameters
- Symmetric key establishment
- Message encoding
- Encryption
- Decryption
- Message decoding

<https://cryptographyacademy.com/des/protocol/>

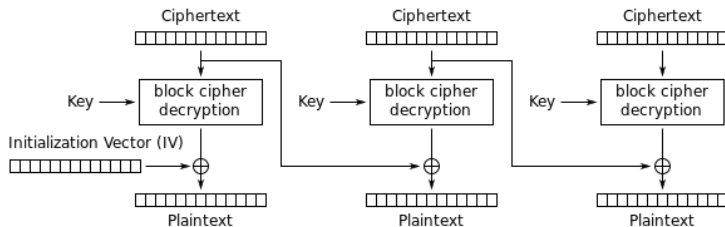
Various modes of operation of Data Encryption Standard (DES)

- Electronic Codebook Mode (ECB)
- Cipher Block Chaining Mode (CBC)
- Cipher Feedback Mode (CFB)
- Output Feedback Mode (OFB)
- Counter Mode (CTR)

Various modes of operation



Cipher Block Chaining (CBC) mode encryption

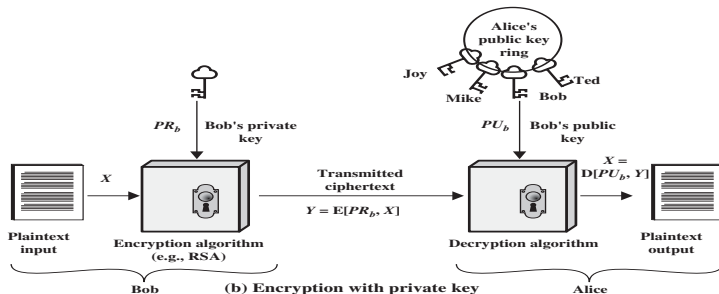
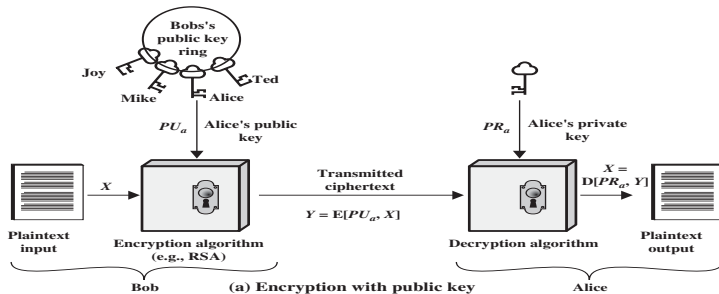


Cipher Block Chaining (CBC) mode decryption

Model of public key encryption

- Consider an encryption scheme consisting of
 - ▶ the set of encryption transformations $\{E_e : e \in K\}$
 - ▶ the set of corresponding decryption transformations $\{D_d : d \in K\}$, where K is the key space.
- The encryption scheme is said to be public-key or asymmetric-key, if for each associated encryption/decryption key pair (e, d) , called public/private key pair, it is computationally “infeasible” to determine private key d from public key e .

Public-Key Cryptography



Introduction

- In 1978, Rivest, Shamir and Adleman at MIT, USA discovered a public-key cryptosystem, known as RSA algorithm.
- They received Turing Award (equivalent to Nobel Prize in Computer Science field).
- Their approach is based on elementary number theory concepts.
- The RSA scheme is a block cipher in which the plaintext and ciphertext are integers between 0 and $n - 1$ for some n . A typical size for n is 1024 bits, or 309 decimal digits.



Figure: Ronald L. Rivest



Figure: Adi Shamir



Figure: Leonard M. Adleman

Key Generation

Table: Key generation of the RSA algorithm

Select p, q	p and q both prime, $p \neq q$ (p and q are large)
Calculate $n = p \times q$	
Calculate $\phi(n) = (p - 1)(q - 1)$ (Euler phi function)	
Select integer e	$\gcd(e, \phi(n)) = 1; 1 < e < \phi(n)$
Calculate d	$d \equiv e^{-1} \pmod{\phi(n)}$
Public key	$KU = \{e, n\}$
Private key	$KR = \{d, n\}$

The RSA Algorithm

Table: Encryption of the RSA algorithm

Plaintext:	$M < n$
Ciphertext:	$C = M^e \pmod{n}$

Table: Decryption of the RSA algorithm

Ciphertext:	C
Plaintext:	$M = C^d \pmod{n}$

Security of the RSA algorithm

- **Brute force:** This involves trying all possible private keys.
- **Mathematical attacks:** There are several approaches, all equivalent in effect to factoring the the product of two primes (Integer Factorization Problem (IFP)).

Given a composite integer n of the form $n = p \times q$, to find the prime factors p and q .

IFP is computationally infeasible (not solvable in polynomial-time factoring algorithm when n is very large, for example, when n is 1024 bits or 2048 bits number.

- **Timing attacks:** These depend on the running time of the decryption algorithm.

The RSA Algorithm

Problem:

The ciphertext message produced by the RSA algorithm with the public key $(e, n) = (223, 1643)$ is:

1451 0103 1263 0560 0127 0897.

Determine the original plaintext message.

Use the standard encoding procedure:

A = 01, B = 02, ..., Z = 26,

, = 27, . = 28, ? = 29,

0 = 30, 1 = 31, ..., 9 = 39, ! = 40,

with 00 as the blank space.

Solution:

- Here $e = 223$, $n = 1643 = 31 \times 53 = p \times q$, say, where p and q are distinct primes.
- $\phi(n) = \phi(1643) = (p - 1) \times (q - 1) = 30 \times 52 = 1560$.
- Using the Extended Euclid's GCD algorithm, $ed \equiv 1 \pmod{\phi(n)}$, that is, $d = 7$.
- The private key is then $(d, n) = (7, 1643)$.
- The given ciphertext blocks are as follows:
 $C_1 = 1451$,
 $C_2 = 0103$,
 $C_3 = 1263$,
 $C_4 = 0560$,
 $C_5 = 0127$
 $C_6 = 0897$.

Solution (Continued...):

- The decipher text (recovered plaintext) of each block C_i is given below (using the repeated square-and-multiply method).
- $M_1 = C_1^d \pmod n = 1451^7 \pmod{1643} = 180$
- $M_2 = C_2^d \pmod n = 103^7 \pmod{1643} = 516$
- $M_3 = C_3^d \pmod n = 1263^7 \pmod{1643} = 122$
- $M_4 = C_4^d \pmod n = 560^7 \pmod{1643} = 500$
- $M_5 = C_5^d \pmod n = 127^7 \pmod{1643} = 141$
- $M_6 = C_6^d \pmod n = 897^7 \pmod{1643} = 523$
- Hence, the original plaintext message using the decoding method given here is as follows:
$$M = M_1 M_2 M_3 M_4 M_5 M_6 = 18\ 05\ 16\ 12\ 25\ 00\ 14\ 15\ 23$$
$$= \text{REPLY NOW}$$

Online Demo on RSA Algorithm

- Generating private/public keys pair
- Encrypting a message
- Decrypting a message

<https://8gwifi.org/rsafunctions.jsp>

https://www.mobilefish.com/services/rsa_key_generation/rsa_key_generation.php